Hall Ticket No: | | | | | | | | | | |

Course Code: A8710

VARDHAMAN

COLLEGE OF ENGINEERING

III B.Tech II Semester Continuous Assessment – I, January - 2026

(Regulations: VCE-R22)

KNOWLEDGE REPRESENTAION AND REASONING (CSM)

Date: 23-01-2026 FN **Time: 2 Hours** **Max. Marks: 30**

Answer all Questions in Part-A

Answer any FOUR Questions in Part-B

Course Outcomes with Bloom's Levels:

CO#	*CO Statement*	*Bloom's Level (L#)*
CO1	Analyse theoretical and practical issues in symbolic knowledge representation and reasoning.	L4
CO2	Apply the capabilities of specific knowledge representation formalisms for specific tasks.	L3
CO3	Make effective use of techniques specific to specific knowledge representation problems and formalisms.	L3
CO4	Make effective use of special-purpose languages for commonsense reasoning, ontologies, planning, reasoning about actions, constraint programming.	L3
CO5	Handle uncertain or incomplete knowledge-based problems.	L4

Questions:

PART-A (Multiple Choice / Fill in the Blanks / Match the Following / Short Answer Type Questions)					
			CO#	*BL#*	*Marks*
1.	a)	State the relationship between knowledge and reasoning.	CO1	L2	**1M**
	b)	What is first-order predicate logic?	CO1	L1	**1M**
	c)	Name any two variety of logic used in AI.	CO1	L1	**1M**
	d)	What are the measures in logical representation?	CO1	L1	**1M**
	e)	How does logic support reasoning processes?	CO1	L2	**1M**
	f)	What is meant by a top-level category?	CO2	L1	***1M***
	g)	What is a physical entity in ontology?	CO2	L1	***1M***
	h)	What is meant by a collection?	CO2	L1	***1M***
	i)	What is the role of time in ontology?	CO2	L2	***1M***
	j)	What is the difference between set and type?	CO2	L2	***1M***

PART-B (Descriptive Questions)				
2.	Discuss the role of logic in knowledge representation and reasoning.	CO1	L3	*5M*
3.	Analyze the role of naming and typing in building scalable knowledge-based systems.	CO1	L4	*5M*
4.	Analyze how types and measures improve the accuracy of logical reasoning.	CO1	L4	*5M*
5.	Explain the concept of types and their role in ontological classification.	CO2	L2	*5M*
6.	Analyze how spatial relationships enhance semantic understanding in ontology.	CO2	L3	*5M*
7.	Analyze the significance of time representation in ontological systems.	CO2	L4	*5M*